

// Day 50 of 60

EVENT STREAMING

Kafka Partitions Explained

→ Swipe through all 6 slides



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The Scalability Wall

When one consumer isn't enough

Topic throughput:
100,000 msgs/second

Single consumer:
10,000 msgs/second

We're 10x behind.
Lag grows forever.

// How do we scale
// horizontally?



Consumers must scale with producers



The Magic of Partitions

A topic isn't one log

Topic 'orders'

Partition 0
Partition 1
Partition 2
Partition 3

```
// Each partition =  
// independent ordered log
```



Partitions are units of parallelism



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Consumer Groups

Distribute partitions across consumers

Group 'order-processor'

Consumer A	← Part 0
Consumer B	← Part 1
Consumer C	← Part 2
Consumer D	← Part 3

```
// Kafka assigns automatically  
// Rebalances on join/leave
```



1 partition → at most 1 consumer per group



Horizontal Scaling Achieved

Linear throughput growth

```
10 partitions  
+ 10 consumers  
= 10x throughput
```

Need more?

Add partitions

Add consumers

Throughput scales linearly

```
// 100K msgs/sec is easy
```

```
// 1M msgs/sec is normal
```



Plan partition count for peak throughput



Ordering Guarantee

Within a partition only

Kafka guarantees:

Order WITHIN a partition

Kafka does NOT guarantee:

Order across partitions

```
// If User A's events go to  
// partition 1, they ALL stay  
// in order – relative to  
// each other.
```



Cross-partition ordering = your job



Routing by Key

Same key → same partition

```
producer.send(  
  topic='orders',  
  key=userId,  
  value=event  
)  
  
hash(userId) % numPartitions  
→ consistent partition  
  
// User A's events always  
// land in same partition  
// → ordering preserved
```



Choose your key carefully



done = true

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